

EVALUATING MOVEMENT QUALITY DURING CUTTING TASKS WITH AI: CAN MOVING THE NEXT GENERATION (MNG) SUPPORT ACL INJURY PREVENTION IN JUNIOR AUSTRALIAN FOOTBALL?

Malishi Nadeema Seena Patabendige¹

¹Institute of Physical Activity and Nutrition, Deakin University

Email: malishiii.nadeema@gmail.com

INTRODUCTION

Australia has among the highest reported rates of anterior cruciate ligament (ACL) reconstruction in the world, with growth increasing most rapidly among 5–14-year-olds [1,2]. Multidirectional sports such as Australian Football (AFL) carry a disproportionate share of this burden, with non-contact side-step cutting specifically identified as a primary ACL injury mechanism, including in the AFLW [3,4]. There is growing interest in using artificial intelligence (AI) to automate movement-quality screening and improve its accessibility beyond elite, well-resourced settings.

Existing field-based qualitative cutting-assessment tools require expert human raters, limiting scalability [5]. Conversely, existing AI-automated screening tools have focused on isolated laboratory tasks such as the drop jump (DJ), automating kinematic variables to output a binary "risk" classification rather than assessing sport-representative movement or identifying targetable technique deficits [6].

This study proposes extending the fundamental dodge movement into Moving the Next Generation (MNG), an existing machine learning-based video assessment platform, to automatically evaluate qualitative side-step cutting technique against established performance criteria [7,8]. Specifically, this study aims to determine whether MNG can reliably evaluate movement quality during cutting tasks in junior AFL players, and whether it identifies ACL-relevant, coachable technique deficits beyond those captured by conventional DJ screening.

METHODS

A cross-sectional, concurrent validity study comparing an AI-automated qualitative assessment of side-step cutting technique (via MNG) against an established DJ based movement quality screen. Twenty-five junior Australian Football players (male and female, aged 10–14 years) will be recruited from community/school-based competitions, excluding those with a current lower-limb injury, prior ACL injury/reconstruction, or any condition affecting movement. Participants will attend a single testing session and complete a standardised DJ protocol (30cm box, three trials) [6,9] and three trials of a side-step cutting task assessed against five established performance criteria [8], filmed from frontal and sagittal views. Cutting quality will be quantified via MNG's automated criteria score [8], and

DJ quality via an automated LESS score [6,9]. Association between the two will be assessed using Spearman's rank correlation and Bland-Altman analysis [10], with divergent cases reviewed qualitatively to identify cutting-specific deficits not reflected in DJ scores.

EXPECTED OUTCOMES

It is hypothesised that MNG-generated cutting scores will show a significant, positive correlation with DJ scores, supporting concurrent validity, with Bland-Altman analysis expected to show acceptable agreement and minimal bias. Some participants are anticipated to pass DJ screening while showing specific cutting deficiencies, demonstrating that MNG captures movement information not reflected in DJ scores.

SIGNIFICANCE

If successful, this study would demonstrate that automated, video-based movement-quality feedback is a viable, low-cost alternative to expert-rater-dependent screening — addressing the scalability barrier limiting ACL injury prevention in community sport. Findings could directly inform how coaches target training, with relevance beyond AFL to other cutting-heavy sports.

REFERENCES

- [1] Zbrojkiewicz D et al. *Med J Aust*. 2018;208(8):354-358.
- [2] Maniar N et al. *Lancet Reg Health West Pac*. 2022;21:100409.
- [3] Rolley TL et al. *Sci Med Football*. 2022;7(2):106-123.
- [4] Cochrane JL et al. *J Sci Med Sport*. 2007;10(2):96-104.
- [5] Dos'Santos T et al. *Phys Ther Sport*. 2019;38:152-161.
- [6] Hébert-Losier K et al. *Appl Sci*. 2020;10(3):892.
- [7] Lander N et al. *J Mot Learn Dev*. 2025;13(3):650-671.
- [8] Victoria Department of Education. *Fundamental motor skills manual*. 1999.
- [9] Padua DA et al. *Am J Sports Med*. 2009;37(10):1996-2002.
- [10] Bland JM, Altman DG. *Lancet*. 1986;327(8476):307-310.